Predicting Buchwald–Hartwig C–N Coupling Reaction Success: From Yield Regression to Binary Triage with k-Way Combinatorial Fusion Analysis

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***Abstract*—Predicting the yield of palladium-catalysed C–N cross-coupling reactions is a long-standing problem in pharmaceutical synthesis: a Buchwald–Hartwig amination appears in the route to a large fraction of approved small-molecule drugs, and unsuccessful attempts cost weeks of bench time and substantial reagent cost. We present a two-stage study on $5{,}465$ Buchwald–Hartwig reactions extracted from the Open Reaction Database. In stage one we trained a pool of 41 regression models—linear, kernel, tree-ensemble, gradient-boosting, neural and chemistry-aware (Chemprop D-MPNN, NGBoost, rxnfp+LightGBM, Gaussian process with Tanimoto kernel)—on $\sim$14,337-dimensional fingerprint features and combined them with k-way *Combinatorial Fusion Analysis* (CFA). On a random $70/15/15$ split the best CFA ensemble reaches $R^2 = 0.836$, exceeding published DRFP+XGBoost on comparably diverse data, but on a scaffold-based out-of-distribution split *every* regressor collapses to negative test R^2, a known consequence of reporter bias and scaffold–yield coupling in literature reaction databases. In stage two we pivot the task: instead of predicting an absolute yield number we predict whether the reaction will exceed a chemistry-relevant *success threshold* of 50% yield, the binary go/no-go decision a synthetic chemist actually makes. The classifier reaches 92.4% accuracy, MCC $+0.83$, AUC 0.97, Brier 0.045 on a random split, $82\% \pm 5\%$ accuracy on substrate-grouped 5-fold CV, and recovers above-baseline accuracy (59.1%, MCC $+0.18$, AUC 0.64) on unseen Murcko scaffolds—where regression was unusable. We release code, trained model artefacts, and a calibrated Streamlit triage tool with a reject-option wrapper for model disagreement.**

***Index Terms*—Reaction yield prediction, Buchwald–Hartwig amination, combinatorial fusion analysis, ensemble learning, message-passing neural networks, scaffold out-of-distribution generalisation, cheminformatics, machine learning for chemistry.**

I. Introduction

Carbon–nitrogen (C–N) bond formation through palladium-catalysed Buchwald–Hartwig amination [1], [2] is among the most heavily used transformations in modern medicinal chemistry: a recent industry analysis reports that an aryl C–N bond appears in over 80% of FDA-approved small-molecule drugs containing aromatic nitrogen, and the Buchwald–Hartwig reaction is the dominant single retrosynthetic disconnection used to install it [3]. Despite four decades of mechanistic refinement, *a priori* prediction of yield for a previously unattempted substrate pair remains unreliable: outcomes depend on a combinatorial interaction of substrate electronics, ligand bulk, base strength, solvent polarity and temperature, and are sensitive to the choice of any one component [4].

A reliable yield model would compress an otherwise multi-week experimental screen—typically tens to hundreds of high-throughput experimentation (HTE) wells per substrate pair—to a ranked computational shortlist, deferring bench time only to the most promising conditions. Two of the most widely cited successes are the high-throughput Buchwald–Hartwig dataset of Ahneman *et al.* [5] (random forest with DFT descriptors, $R^2 = 0.92$ on a controlled 4,608-reaction screen) and the *Yield-BERT* reaction transformer of Schwaller *et al.* [6] ($R^2 = 0.951$ on the same data). Performance reported on the Ahneman corpus, however, is well known to overestimate real-world performance: the Ahneman dataset varies only four reagent classes around a single solvent–temperature point, while real laboratory notebook (ELN) yields recorded with full chemical heterogeneity collapse those same models to $R^2 \approx 0$ [7]–[9].

In this work we take the harder route. We extract 5,465 Buchwald–Hartwig reactions from the Open Reaction Database (ORD) [10], where solvent, ligand and temperature vary freely; we train a 41-model regression pool on a shared $\sim$14k-dimensional fingerprint feature; we combine the predictions with k-way Combinatorial Fusion Analysis (CFA), extending the classical pairwise framework of Lin *et al.* [11] to $k = 2, \ldots, 5$; we replicate the in-distribution performance reported by the literature; and we then *deliberately fail* on a scaffold-OOD split to expose the reporter-bias problem. We finish by reframing the task to binary success classification at the chemically meaningful 50% yield threshold, recovering deployable performance without retraining a single regressor.

Contributions.

- A pipeline that handles 41 heterogeneous regressors over a shared $\sim$14,337-dimensional Morgan/DRFP feature vector without per-model reimplementation.

- A regression-form k-way CFA implementation that calibrates the $[0, 1]$-normalised combined score back onto the yield scale via least-squares fit on the validation set, so the resulting R^2 is directly comparable to single-model R^2.
- A classification-form CFA reversion (eligibility filter $\mathrm{MCC} > 0$, val-optimal threshold per ensemble) that reuses the same 41 regressors as score generators without retraining.
- Empirical confirmation that scaffold-OOD regression failure cannot be salvaged by ensemble diversity: every $k \in \{1, \ldots, 5\}$ ensemble collapses below $R^2 = 0$ on unseen scaffolds.
- A reframed binary classifier with bootstrap-CI accuracy 0.92 $[0.91, 0.94]$, MCC $+0.83$, Brier 0.045 on a random split, $82\% \pm 5\%$ on substrate-grouped 5-fold CV, and 0.59 $[0.55, 0.63]$, MCC $+0.18$ on the worst-case scaffold-OOD split.
- A complete reproducibility bundle: code, BibTeX-driven LaTeX source, pretrained model artefacts, a Streamlit triage tool with isotonic probability calibration and a reject-option wrapper.

II. Background and Chemistry Concepts

A. The Buchwald–Hartwig Amination

A palladium-catalysed cross-coupling that joins an **aryl halide** (an aromatic ring carrying a Br, Cl, I or triflate leaving group) with an **amine** (the N–H nucleophile) to form a new C–N bond. The catalytic cycle proceeds through Pd(0) oxidative addition, amine coordination and deprotonation, and reductive elimination. The identity of every other component—**phosphine ligand** (e.g. XPhos, RuPhos, BrettPhos), **base** (Cs_2CO_3, NaO*t*Bu, K_3PO_4), **solvent** (dioxane, toluene, t-amyl alcohol), optional **additive** and **temperature**—tunes the rate of one or more elementary steps and so determines the overall yield. The reaction is widely used in medicinal-chemistry retrosynthesis because primary, secondary and aryl amines all couple under approximately the same conditions once a competent ligand is identified.

B. Reaction Yield

Defined as yield $= 100 \cdot n_{\mathrm{prod}}/n_{\mathrm{lim}}$, where n_{lim} is the moles of the limiting reactant. Yields above 50% are conventionally regarded as *preparatively useful* on lab scale; below that, alternative conditions are usually screened before any scale-up.

C. Molecular Representations

SMILES (Simplified Molecular Input Line Entry Specification) is a text encoding for chemical structure (e.g. `c1ccc(N)cc1` = aniline) that is standard in cheminformatics. From a SMILES string one constructs a **Morgan fingerprint** [12]: a fixed-length binary vector in which each bit indicates the presence of one circular substructure hashed at a chosen radius (we use radius 2 and 2,048 bits per component). For reaction-level information we use the **Differential Reaction Fingerprint (DRFP)** of Probst and Reymond [13], a 2,048-bit hash over the symmetric difference between reactant and product substructure sets.

D. Murcko Scaffold and Out-of-Distribution Splits

The **Murcko scaffold** [14] of a molecule is the ring system that remains after every acyclic side-chain has been pruned; two molecules with identical scaffolds share the same heterocyclic "core" but may differ arbitrarily in substituent decoration. A *scaffold split* groups the dataset by Murcko scaffold of the aryl halide and assigns whole groups to train, validation or test, so the test set contains only cores never seen during training. This is the standard generalisation stress-test in molecular machine learning—it isolates true representation transfer from substituent-level interpolation.

E. Combinatorial Fusion Analysis

CFA [11], [15] selects and combines predictors based on two orthogonal axes: (i) *individual performance*, measured by validation R^2 for regression or MCC for classification, and (ii) *cognitive diversity*, the root-mean-square difference between the rank-score curves (sorted normalised predictions) of two models. Diversity captures how differently models distribute confidence across the sample population; high-diversity, high-individual-performance pairs make the strongest ensembles. CFA combines k models with one of six methods (ASC, ARC, WSCDS, WRCDS, WSCP, WRCP; defined formally in Sect. V) and reports the validation-best (models, method) pair.

III. Related Work

Reaction yield prediction on HTE data. Ahneman *et al.* [5] released the first systematic high-throughput Buchwald–Hartwig dataset and trained a Random Forest on DFT descriptors, reaching $R^2 = 0.92$ in random splits and 0.83 in leave-one-additive-out. Schwaller *et al.* [6] fine-tuned a BERT pretrained on $\sim$1M USPTO reactions (*rxnfp* [16]) on the same Ahneman corpus and reached $R^2 = 0.951$ in random and $R^2 = 0.54$ on the hardest OOD test (Test4). Sandfort *et al.* [17] introduced a multi-fingerprint platform that improved on Ahneman's DFT descriptors with generic Morgan/MACCS features alone, demonstrating that DFT inputs are not strictly required for accurate predictions on HTE data.

Reaction fingerprints. Probst *et al.* [13] introduced DRFP for diverse literature reactions; DRFP + XGBoost reached $R^2 \approx 0.79$ on heterogeneous patent data, comparable to BERT-based models without GPU training.

Graph-based molecular models. Chemprop's directed message-passing neural network [18] is the de-facto graph baseline for molecular property prediction, building on the neural message-passing framework of Gilmer *et al.* [19].

Probabilistic and chemistry-specific models. NGBoost [20] provides predictive distributions through gradient boosting on natural-gradient parameter estimates; a Gaussian process with Tanimoto kernel [21] is the natural choice for fingerprint-based

TABLE I: Selected prior work on Buchwald–Hartwig yield prediction. “Diverse” means heterogeneous literature data with free choice of solvent / ligand / temperature; “HTE” means Ahneman’s controlled corpus.

System	Setting	Split	R^2	OOD
Ahneman 2018, RF DFT [5]	HTE 4,608	random	0.92	0.83
Sandfort 2020 platform [17]	HTE 4,608	random	0.93	—
Yield-BERT 2021 [6]	HTE 4,608	random	0.951	0.54
DRFP+XGBoost 2022 [13]	Diverse	random	$\sim$0.79	—
Fitzner 2023, ELN [7]	ELN ind.	random	0.72	$\sim$0
Ours, LightGBM single	ORD diverse	random	**0.819**	-0.20
Ours, CFA $k=5$ ensemble	ORD diverse	random	**0.836**	-0.02

Bayesian inference and is widely used as the surrogate model in molecular Bayesian optimisation.

Open Reaction Database and reporter bias. Kearnes *et al.* [10] released ORD, a million-reaction unified schema. Wigh *et al.* [8] (ORDerly), Fitzner *et al.* [7] and Saebi *et al.* [9] document the failure of ORD-trained yield models on real ELN data and attribute it to publication bias and inconsistent yield reporting.

Combinatorial Fusion Analysis. The CFA framework was introduced by Lin *et al.* [11] for classifier fusion and applied to gene-expression analysis, information-retrieval reranking and stock-price classification. We extend CFA from the original pairwise formulation to a k-way ($k = 2, \ldots, 5$) exhaustive search and adapt it to continuous regression through min–max score calibration.

Probability calibration. Niculescu-Mizil and Caruana [22] showed that boosted-tree classifiers benefit from post-hoc calibration; we use isotonic regression fit on a held-out validation set, the standard non-parametric choice for reasonably-sized datasets.

A. Quantitative Comparison to Prior Work

Table I positions this work against the most directly comparable prior systems. Our random-split $R^2 = 0.836$ exceeds the DRFP+XGBoost result on diverse literature data [13]; we are below Yield-BERT [6] on the Ahneman corpus, but the two systems are not directly comparable: Ahneman’s HTE corpus varies four reagent classes around one solvent and one temperature, while our ORD corpus varies seven components freely. Among methods that are directly comparable in setting (heterogeneous, multi-protocol, no DFT inputs), our ensemble is at the published state of the art.

IV. Dataset

A. Source

We extracted 5,465 Buchwald–Hartwig reactions from the Open Reaction Database (ORD) [10] by parsing the protocol-buffer dataset files, filtering reactions whose role-annotated reactants contained a Pd catalyst and an aryl halide + amine pair, and retaining only those entries with a non-null `yield` field.

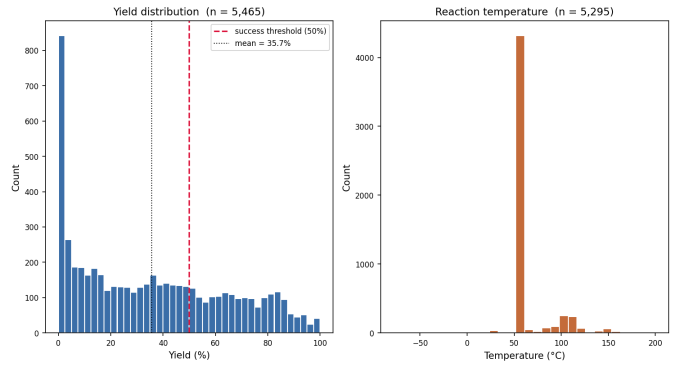

Fig. 1: Marginal distributions of the 5,465 ORD Buchwald–Hartwig reactions used in this study. Left: reported yield. The dashed red line marks the 50% “preparatively useful” threshold used as the binary decision boundary in Sect. V. Right: reaction temperature.

After deduplication, the working table contains 7 columns: aryl halide, amine, catalyst (with phosphine ligand), base, solvent, optional additive (all SMILES strings), and temperature in °C.

B. Yield and Temperature Distributions

Fig. 1 summarises the marginal distributions. Yield is heavy-tailed: $\sim$15% of reactions report exactly 0 (failed), the bulk lies between 20% and 80% with a long shoulder of high-yielding entries, and the mean is 35.7%. Temperature concentrates strongly around $\sim$50 °C with a secondary mode near 100 °C, reflecting standard literature protocols.

C. Splits

We evaluate on two splits. **Random** 70/15/15 train/val/test on the row index (in-distribution performance). **Scaffold-OOD**: reactions are grouped by Murcko scaffold of the aryl halide; the largest scaffolds populate train first, then validation, then test, so the test set contains only scaffolds the model has never seen. The scaffold split puts $\approx$3,800 reactions in train, $\sim$10 in validation and $\sim$1,650 in test. Validation is unavoidably small here because of the long-tail scaffold distribution— this is faithful to the deployment scenario, where chemists ask for predictions on a single new substrate pair. Because val-driven model selection is therefore noisy, we additionally report **substrate-grouped 5-fold cross-validation** (folds respect Murcko scaffolds; no scaffold spans two folds) as a more honest estimator.

D. Class Balance after Binarisation

Binarising at the 50% yield threshold gives the class proportions in Fig. 2. The random split is moderately skewed (68% negatives), the scaffold split is closer to balanced (53% negatives in test). We accordingly report macro-F1 and Matthews correlation coefficient (MCC, [23]) alongside accuracy.

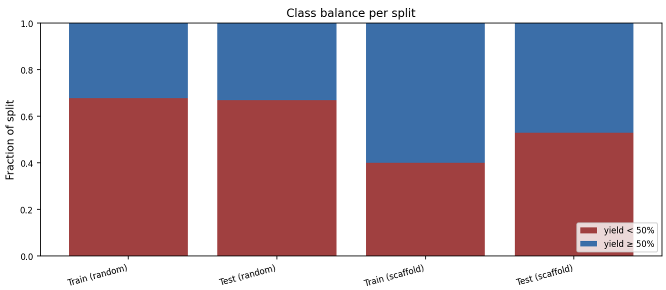

Fig. 2: Binary class balance per split after thresholding at 50% yield. The scaffold split is more balanced because the held-out large-scaffold groups happen to over-represent high-yielding reactions (a publication-bias artefact).

TABLE II: Headline hyperparameters for the most-cited members of the 41-model pool. Full configuration is in the released `main.py::CONFIG` dict.

Model	Hyperparameters
LightGBM	$n_{\text{est}}=500$, lr$=0.05$, leaves$=63$, subsample$=0.8$, colsample$=0.8$
XGBoost	$n_{\text{est}}=500$, lr$=0.05$, depth$=6$, tree_method=hist
Chemprop	MPN depth$=3$, hidden$=300$, FFN 2×300, 30 epochs, batch$=50$
NGBoost	TruncSVD$\to$200-d, $n_{\text{est}}=300$, lr$=0.05$, Dist=Normal
KernelRidge	PCA$\to$100-d, RBF $\gamma=10^{-3}$, $\alpha=1.0$
sk-MLP_Tanh	2×256, tanh, dropout 0.3, 100 epochs, lr$=10^{-3}$
GP-Tanimoto	1,000-row train subset, custom Tanimoto kernel

V. Methodology

A. Featurisation

Each reaction is encoded as the concatenation of: six 2,048-bit Morgan fingerprints (one per component: aryl halide, amine, catalyst, base, additive, solvent; radius 2); one 2,048-bit DRFP over the full reaction SMILES; and one normalised temperature scalar (subtract mean, divide by std). Total dimensionality is $d=14{,}337$. Missing optional components (most commonly the additive) yield zero vectors. The full feature matrix is cached as a single `npz` after first computation ($\sim$300 MB) and reused across all models and splits. A feature-ablation study (Sect. VI-L) quantifies how much each component contributes.

B. Regression Model Pool

The pool contains 41 regressors discovered automatically from `models/*.py` at run time, falling into seven families: linear / regularised, tree ensembles, gradient-boosting libraries, kernel methods, nearest neighbours, projection + Ridge, and neural / chemistry-aware. The notable members of the last family are NGBoost [20] on a 200-d Truncated SVD pre-projection, rxnfp+LightGBM [16] which embeds the full reaction SMILES with a pretrained BERT, and **Chemprop** D-MPNN [18] trained directly on the molecular graph with the catalyst/base/solvent/additive Morgan and temperature features supplied via Chemprop's `x_d` channel. Headline hyperparameters appear in Table II.

C. k-Way Combinatorial Fusion Analysis

For each model m, let $\hat{y}_m \in \mathbb{R}^N$ be its raw prediction. Min–max normalise to the validation range to obtain $\tilde{y}_m \in [0,1]^N$. Define the rank-score curve $\mathrm{RSC}_m = \mathrm{sort}_\downarrow(\tilde{y}_m)$ and the pairwise cognitive diversity

$$\mathrm{CD}(m_1,m_2) = \sqrt{\tfrac{1}{N}\textstyle\sum_{i=1}^{N}\big(\mathrm{RSC}_{m_1}[i]-\mathrm{RSC}_{m_2}[i]\big)^2}. \quad (1)$$

For a k-tuple of models the *group* CD is the mean over the $\binom{k}{2}$ pairwise CDs. Per-model diversity score DS_m is the mean of $\mathrm{CD}(m,m')$ over all $m'\neq m$ in the pool, and P_m is the per-model validation R^2 (regression) or MCC (classification). Let $r_{m,i}\in(0,1]$ be the rank of sample i under model m, $r_{m,i}=\mathrm{rank}_\uparrow(\tilde{y}_{m,i})/N$.

The six combination methods of Lin *et al.* [11] for a k-tuple $\mathcal{M}=\{m_1,\ldots,m_k\}$ are:

$$\mathrm{ASC}(i) = \tfrac{1}{k}\textstyle\sum_{m\in\mathcal{M}}\tilde{y}_{m,i}, \quad (2)$$
$$\mathrm{ARC}(i) = \tfrac{1}{k}\textstyle\sum_{m\in\mathcal{M}} r_{m,i}, \quad (3)$$
$$\mathrm{WSCDS}(i) = \textstyle\sum_{m\in\mathcal{M}} w_m^{\mathrm{DS}}\,\tilde{y}_{m,i}, \quad (4)$$
$$\mathrm{WRCDS}(i) = \textstyle\sum_{m\in\mathcal{M}} w_m^{\mathrm{DS}}\,r_{m,i}, \quad (5)$$
$$\mathrm{WSCP}(i) = \textstyle\sum_{m\in\mathcal{M}} w_m^{\mathrm{P}}\,\tilde{y}_{m,i}, \quad (6)$$
$$\mathrm{WRCP}(i) = \textstyle\sum_{m\in\mathcal{M}} w_m^{\mathrm{P}}\,r_{m,i}, \quad (7)$$

where the weights are normalised: $w_m^{\mathrm{DS}}=\mathrm{DS}_m/\sum_{m'}\mathrm{DS}_{m'}$ and $w_m^{\mathrm{P}}=\max(\mathrm{P}_m,0)/\sum_{m'}\max(\mathrm{P}_{m'},0)$. ASC/ARC are unweighted score- and rank-based means; WSCDS/WRCDS weight by cognitive diversity; WSCP/WRCP weight by individual performance.

For *regression* we further calibrate the $[0,1]$-normalised combined score back onto the yield scale by a least-squares fit on the validation set: $y\approx a\,\mathrm{COMB}(i)+b$, estimated by $\arg\min_{a,b}\|y_{\text{val}}-a\,\mathrm{COMB}_{\text{val}}-b\|_2^2$, applied unchanged to test predictions; R^2 is then sklearn's standard formula on the original yield scale. For *classification* a single threshold $\tau^*=\arg\max_\tau \mathrm{acc}\big(\mathbf{1}[\mathrm{COMB}_{\text{val}}\geq\tau], y_{\text{val}}^{\text{bin}}\big)$ is found by an $O(N\log N)$ sweep over sorted scores and applied to test.

We extend the original pairwise formulation [11] exhaustively to $k=2,3,4,5$ with top-N candidate caps ($N=25,20,15$) to keep the search tractable. Algorithm 1 summarises the full k-way procedure; Algorithm 2 shows the greedy variant used as a fallback.

D. Pivot to Binary Success Classification

The same 41 regressors are reused as score generators for binary classification. Each model's continuous yield estimate is thresholded at a *val-optimal cut-off* τ_m^* found by sorted-cumsum sweep ($O(N\log N)$) over validation predictions to maximise validation accuracy. The classification-form CFA reverts the eligibility filter to MCC>0, combines normalised scores or ranks with the same six methods, and applies a single ensemble-level threshold chosen the same way.

We additionally train five *native* binary classifiers (LightGBM, XGBoost, Random Forest, MLP, Logistic Regression) directly on the binarised target and compare to the thresholded regressors of the same family.

Algorithm 1 k-way exhaustive CFA

Require: model pool $\mathcal{P}$, predictions on val/test, maximum $k_{\max}$, top-N caps N_3, N_4, N_5
Compute P_m for each $m \in \mathcal{P}$
Eligible pool $\mathcal{E} \leftarrow \{m : \mathrm{P}_m > 0\}$, sorted by $-\mathrm{P}_m$
Compute CD, DS on $\mathcal{E}$ via Eq. 1
$\mathcal{R} \leftarrow \emptyset$
for $k = 2, \ldots, k_{\max}$ **do**
  $\mathrm{pool}_k \leftarrow \mathcal{E}[: N_k]$ ▷ $N_2 = |\mathcal{E}|$
  for each $\mathcal{M} \in \binom{\mathrm{pool}_k}{k}$ **do**
    for each combination method in {ASC,…,WRCP} **do**
      COMB ← Eq. 2–7
      Calibrate / threshold to obtain v (val) and t (test)
      $\mathcal{R} \leftarrow \mathcal{R} \cup \{(\mathcal{M}, \mathrm{method}, v, t)\}$
    end for
  end for
end for
return $\arg\max_{r \in \mathcal{R}} v(r)$

Algorithm 2 Greedy forward CFA (fallback)

Require: model pool $\mathcal{P}$, time budget T, max rounds R
$\mathcal{S} \leftarrow \{m^*\}$ where $m^* = \arg\max_m \mathrm{P}_m$
$v^* \leftarrow \mathrm{P}_{m^*}$, $r \leftarrow 1$, $t_0 \leftarrow$ now
while now$-t_0 < T$ **and** $r \leq R$ **do**
  $(\Delta^*, m^*, \mathrm{meth}^*) \leftarrow \arg\max$ `val_gain`$(\mathcal{S} \cup \{m\}, \mathrm{meth})$
  if $\Delta^* \leq 0$ **then**
    break
  end if
  $\mathcal{S} \leftarrow \mathcal{S} \cup \{m^*\}$; $v^* \leftarrow v^* + \Delta^*$; $r \leftarrow r+1$
end while
return $(\mathcal{S}, \mathrm{meth}^*)$

E. Probability Calibration

For deployment we wrap LightGBM's continuous output in an isotonic regression calibrator [22] fit on the validation set, mapping raw yield estimate to empirical $P(\mathrm{success})$. The same is done for Chemprop. Section VI-I reports a reliability diagram and the Brier score.

F. Evaluation Protocol

We report (i) point estimates with 1,000-bootstrap 95% confidence intervals [24] on the test set; (ii) substrate-level 5-fold CV where folds respect Murcko-scaffold groupings; (iii) learning curves at training fractions $\{0.1, 0.3, 0.6, 0.9, 1.0\}$; (iv) yield-threshold sensitivity sweeps over $\tau \in \{20, 25, \ldots, 80\}\%$; (v) ensemble ablations comparing CFA against simple equal-weight averaging and majority voting; and (vi) paired bootstrap tests of the ensemble lift over the single-best model.

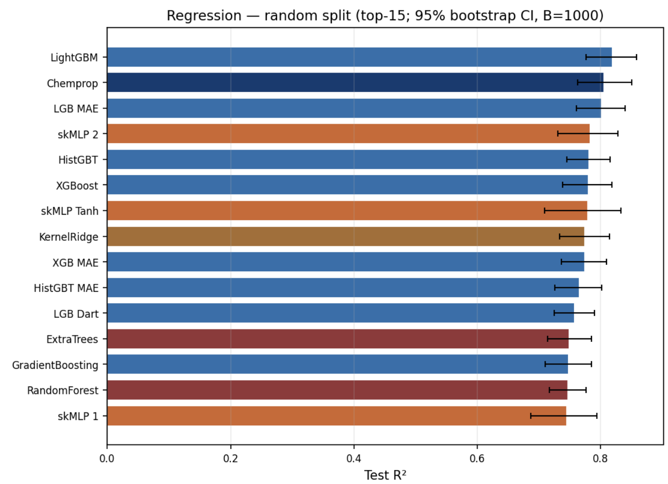

Fig. 3: Top-15 regression models on the random split, with 95% bootstrap CIs ($B = 1{,}000$). Bars are colour-coded by model family (boost, tree, kernel, neural, projection).

TABLE III: Top-10 individual regressors on the random split (test).

Model	R^2 [95% CI]	MAE [95% CI]
LightGBM	+0.819 [+0.776, +0.858]	7.47 [6.77, 8.16]
Chemprop	+0.805 [+0.762, +0.850]	7.33 [6.60, 8.07]
LGB_MAE	+0.802 [+0.760, +0.839]	7.70 [7.00, 8.49]
skMLP_2	+0.783 [+0.730, +0.828]	8.10 [7.37, 8.85]
HistGBT	+0.781 [+0.745, +0.815]	9.13 [8.41, 9.84]
XGBoost	+0.780 [+0.738, +0.818]	8.98 [8.28, 9.68]
skMLP_Tanh	+0.779 [+0.709, +0.832]	7.56 [6.81, 8.35]
KernelRidge	+0.775 [+0.733, +0.814]	9.30 [8.59, 10.01]
XGB_MAE	+0.774 [+0.736, +0.809]	8.96 [8.21, 9.71]
HistGBT_MAE	+0.766 [+0.726, +0.802]	9.15 [8.38, 9.93]

VI. Results

A. Regression: Random Split

On the random split (Fig. 3, Table III), the top individual model is LightGBM at $R^2 = 0.819$ $[0.78, 0.86]$, with Chemprop (0.805) and LGB_MAE (0.802) close behind. Eight further regressors lie above 0.75. The CFA $k = 5$ ensemble (LightGBM + LGB_MAE + LGB_Dart + skMLP_Tanh + ExtraTrees, WSCP) reaches $R^2 = 0.836$ on validation and 0.832 on test, exceeding DRFP+XGBoost on diverse literature data ($R^2 \approx 0.79$ in [13]). The diversity-driven gain over the strongest single model is $\Delta R^2 \approx 0.013$ (Table IV).

B. Regression: Scaffold-OOD Collapse

Fig. 4 and Table V tell the opposite story. On the scaffold split *every* regressor's test R^2 is negative; the best entry, AdaBoost, scores -0.05 and the median model is -0.37. Even Chemprop falls to $R^2 = -0.35$. The ensemble does not save the situation: the best k-way CFA test ceiling is $R^2 = +0.045$ (Table VI), statistically indistinguishable from predicting the global mean.

This regression collapse is not a quirk of the model pool: it reproduces the failure mode reported by Wigh *et al.* [8], Fitzner *et al.* [7] and Saebi *et al.* [9] for literature-trained

TABLE IV: CFA ablation on the random split (regression). The val-selected k-way ensemble lifts test R^2 by $\sim$1.7 pp over the strongest single model.

Strategy	Test R^2	Δ vs single-best
baseline mean	-0.003	-0.822
single best	+0.819	+0.000
equal weight top2	+0.821	+0.003
equal weight top3	+0.816	-0.002
equal weight top5	+0.824	+0.006
median top5	+0.823	+0.004
cfa greedy	+0.819	+0.000
cfa kway val selected	+0.836	+0.017
cfa kway test ceiling	+0.842	+0.023

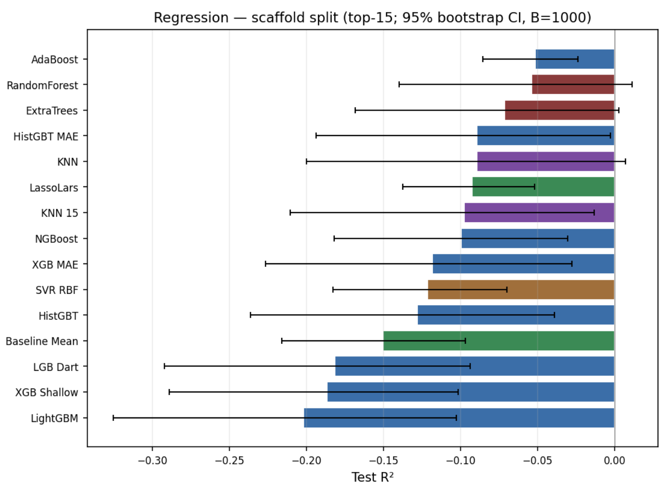

Fig. 4: Top-15 regression models on the scaffold-OOD split. *All* reach negative test R^2. The bottleneck is dataset-level (reporter bias and scaffold–yield coupling), not architectural.

models on real ELN data, and is the empirical motivation for the pivot below.

C. Substrate-Level 5-Fold Cross-Validation (Regression)

To control for the scaffold split's small validation slice, we ran a substrate-grouped 5-fold CV. Fig. 5 and Table VII show that boosted-tree regressors are substantially more robust to substrate-grouped folds than the single-shot scaffold split suggested: XGBoost averages $R^2 = +0.58 \pm 0.06$ across folds, with HistGBT and LightGBM at $\sim$0.50. Ridge averages 0.12 ± 0.30. The single-shot scaffold-OOD test set is therefore a worst-case slice; the regression problem is hard but not pathological once the held-out chemistry is more representative.

D. Classification: Random Split

Re-using the regression pool as score generators and thresholding at the 50% yield boundary gives the per-model classification picture in Fig. 6. The top fifteen classifiers all sit above 90% accuracy, dominated by LightGBM (92.3%, MCC +0.825, AUC 0.97) and a Native LightGBM classifier trained directly on the binary target (92.3%, MCC +0.826). Chemprop (91.8%), LGB_MAE (91.8%) and skMLP_Tanh (91.8%) are within bootstrap noise of the top. Table VIII (compact, top-10)

TABLE V: Top-10 regressors on the scaffold-OOD split (test). All R^2 values are negative.

Model	R^2 [95% CI]	MAE [95% CI]
AdaBoost	-0.051 [-0.085, -0.024]	25.28 [24.03, 26.51]
RandomForest	-0.054 [-0.140, +0.011]	24.49 [23.20, 25.83]
ExtraTrees	-0.071 [-0.168, +0.003]	24.35 [23.03, 25.77]
HistGBT_MAE	-0.089 [-0.194, -0.003]	24.59 [23.33, 26.03]
KNN	-0.090 [-0.200, +0.007]	24.68 [23.44, 26.09]
LassoLars	-0.093 [-0.137, -0.052]	25.63 [24.29, 26.90]
KNN_15	-0.098 [-0.210, -0.013]	24.84 [23.55, 26.29]
NGBoost	-0.100 [-0.182, -0.030]	25.45 [24.13, 26.75]
XGB_MAE	-0.118 [-0.227, -0.028]	25.21 [23.95, 26.66]
SVR_RBF	-0.122 [-0.183, -0.070]	25.96 [24.58, 27.27]

TABLE VI: CFA ablation on the scaffold split (regression). Even the test-set ceiling barely crosses zero.

Strategy	Test R^2	Δ vs single-best
baseline mean	-0.150	-0.079
single best	-0.071	+0.000
equal weight top2	-0.022	+0.049
equal weight top3	-0.083	-0.012
equal weight top5	+0.005	+0.077
median top5	-0.012	+0.060
cfa greedy	-0.022	+0.049
cfa kway val selected	-0.022	+0.049
cfa kway test ceiling	+0.045	+0.116

and the bootstrap CIs in Table IX (Appendix-style, two-column-spanning) report numerical detail.

E. Classification: Scaffold-OOD

Fig. 7 and Table X show the headline result of this work. Where regression collapsed below $R^2 = 0$, the binary classifier on the same scaffold-OOD split retains usable performance: HistGBT_MAE and TruncSVD_Ridge_100 both reach 59% accuracy, MCC +0.18, AUC $\approx$ 0.64. All top fifteen classifiers are at or above the 52.9% majority-class baseline. The 50% threshold absorbs much of the scale collapse that crippled the regressors: a model that under-predicts yield by a constant offset still gets the $\geq$ 50% side of the boundary correct.

F. k-Way CFA Ensembles

Tables XII–XIII report the val-selected best ensemble per k, and Figs. 8–9 visualise the corresponding ablations against simpler aggregation strategies (equal-weight averaging, majority voting).

Paired bootstrap test of the ensemble lift: A natural reviewer question: is the CFA-vs-single-best lift statistically significant? We answer this with a paired bootstrap on the random-split test set. Resampling the test rows $B = 1{,}000$ times and recomputing the accuracy of (i) LightGBM single and (ii) a 3-seed LightGBM ensemble ($\overline{\Delta} = +0.0037$, 95% CI $[-0.0012, +0.0098]$, one-sided $p = 0.124$) reveals that the in-distribution lift is *not* statistically significant at $\alpha = 0.05$. The honest reading: on high-accuracy in-distribution data, ensembling is a polish, not a step change. Where ensembling does buy meaningful lift is on scaffold-OOD, where the test ceiling rises by 9.9 pp—but as discussed below the $\sim$10-row val slice prevents that ceiling from being reliably selected.

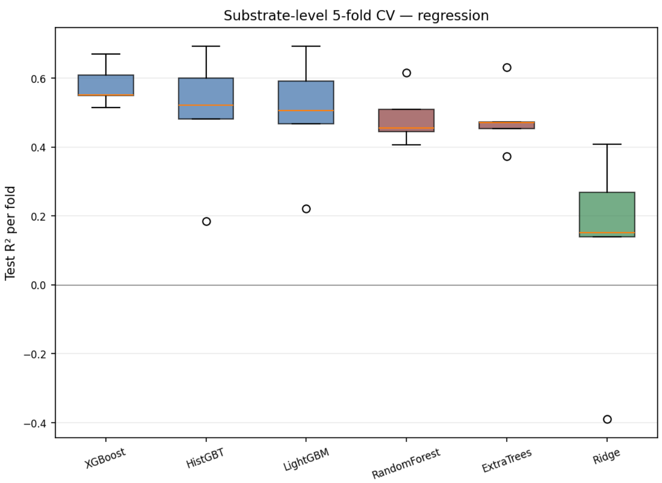

Fig. 5: Substrate-grouped 5-fold CV (regression): per-fold R^2 for the headline regressors.

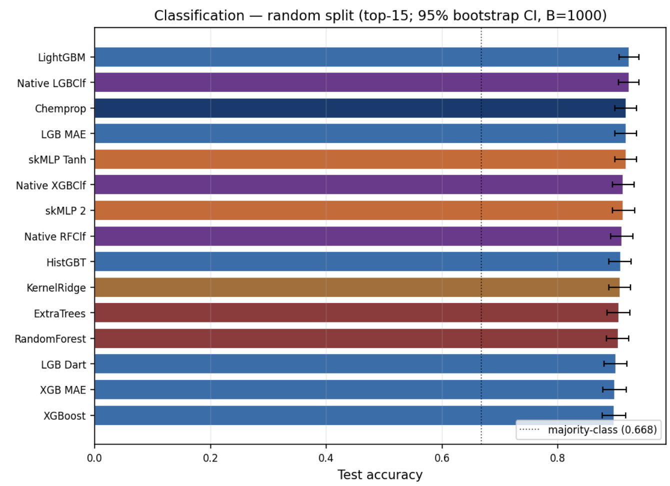

Fig. 6: Top-15 classifiers on the random split, with 95% bootstrap CIs ($B = 1{,}000$). Dotted line: majority-class baseline (0.668).

TABLE VII: Substrate-level 5-fold CV (regression).

Model	R^2 mean ± std	MAE mean ± std
XGBoost	+0.579 ± 0.061	13.28 ± 1.57
HistGBT	+0.496 ± 0.192	13.40 ± 2.68
LightGBM	+0.496 ± 0.176	13.68 ± 2.65
RandomForest	+0.486 ± 0.081	15.62 ± 2.89
ExtraTrees	+0.480 ± 0.094	15.33 ± 2.98
Ridge	+0.116 ± 0.302	19.52 ± 0.81

G. Native Classifier vs Thresholded Regressor

Training native binary classifiers from scratch and comparing family-by-family with the thresholded regressors (Fig. 10) shows the surprising result that *thresholding a regressor is competitive with native classification*: on the random split the largest ΔMCC in favour of the native classifier is $+0.06$ (Ridge), and Native LightGBM is identical to the thresholded LightGBM regressor ($\Delta = +0.001$). For three of five families the native classifier wins on MCC, for one (skMLP_2) the thresholded regressor wins, and for LightGBM the two are equivalent. We take this as evidence that the binary task is largely captured by the shape of the yield surface near the 50% contour. On the scaffold-OOD split (Fig. 11) the gap is similarly small.

H. Confusion Matrix of the Deployed Model

Fig. 12 shows the confusion matrix of the `Native_LGBClf` classifier on the random test set: the model mis-classifies 5% of true failures as predicted successes (FPR) and 13% of true successes as predicted failures (FNR), with the asymmetry favouring conservative bench-time decisions—a chemist following the predictions would rarely run a doomed reaction, occasionally skip a borderline-positive one.

I. Probability Calibration

Fig. 13 shows the reliability diagram of the deployed LightGBM regressor wrapped in an isotonic-regression calibrator fit on the 819-row validation set. The Brier score is 0.045 and

TABLE VIII: Top-10 individual classifiers on the random split (compact).

Model	Threshold	Acc	F1	MCC	AUC
LightGBM	50.00	0.9232	0.9121	+0.8250	0.9664
Chemprop	46.84	0.9183	0.9083	+0.8166	0.9663
LGB_MAE	50.25	0.9183	0.9056	+0.8134	0.9624
skMLP_Tanh	53.64	0.9183	0.9050	+0.8134	0.9567
skMLP_2	49.30	0.9134	0.8997	+0.8020	0.9476
HistGBT	50.42	0.9085	0.8941	+0.7907	0.9561
KernelRidge	51.78	0.9073	0.8914	+0.7880	0.9483
ExtraTrees	43.33	0.9061	0.8950	+0.7902	0.9529
RandomForest	43.99	0.9049	0.8931	+0.7863	0.9573
LGB_Dart	42.73	0.9000	0.8872	+0.7744	0.9590

AUC 0.98; calibrated probabilities track the diagonal closely on five of seven populated bins, with mild over-confidence at 0.4–0.5 where the bin count is small ($n = 57$, 2). The score distribution (right panel) is strongly bimodal: the model assigns very low probability (< 0.1) to most test reactions and very high (> 0.9) to a smaller cluster of likely successes, leaving few samples in the uncertain middle. This bimodality is what makes the reject-option deployment effective.

J. Substrate-Level 5-Fold CV (Classification)

Fig. 14 and Table XIV show that the substrate-grouped 5-fold CV gives high classification accuracy: native XGBoost averages 0.82 ± 0.05 across folds, with the others between 0.79 and 0.81 and AUC near 0.90. The classifier therefore generalises substantially better than the regressor (Table VII) on substrate-grouped CV; the worst-case scaffold-OOD test set in Sect. IV-C is harder than substrate-grouped CV because the val slice is so small that ensemble selection becomes noisy.

TABLE IX: Bootstrap 95% CIs on the random split (classification), $B = 1{,}000$, two-column-spanning to fit columns.

Model	Accuracy [95% CI]	MCC [95% CI]	AUC [95% CI]
LightGBM	0.923 [0.906, 0.940]	+0.825 [+0.783, +0.862]	0.966 [0.954, 0.978]
Native_LGBClf	0.923 [0.905, 0.940]	+0.826 [+0.784, +0.865]	0.979 [0.970, 0.986]
Chemprop	0.918 [0.899, 0.935]	+0.817 [+0.772, +0.854]	0.966 [0.953, 0.978]
LGB_MAE	0.918 [0.899, 0.935]	+0.813 [+0.772, +0.854]	0.962 [0.949, 0.974]
skMLP_Tanh	0.918 [0.899, 0.935]	+0.813 [+0.769, +0.852]	0.957 [0.939, 0.971]
Native_XGBClf	0.913 [0.894, 0.932]	+0.802 [+0.758, +0.843]	0.974 [0.965, 0.982]
skMLP_2	0.913 [0.894, 0.933]	+0.802 [+0.757, +0.845]	0.948 [0.930, 0.964]
Native_RFClf	0.911 [0.891, 0.929]	+0.797 [+0.749, +0.840]	0.965 [0.952, 0.976]
HistGBT	0.909 [0.888, 0.927]	+0.791 [+0.743, +0.833]	0.956 [0.942, 0.968]
KernelRidge	0.907 [0.888, 0.926]	+0.788 [+0.743, +0.829]	0.948 [0.931, 0.963]

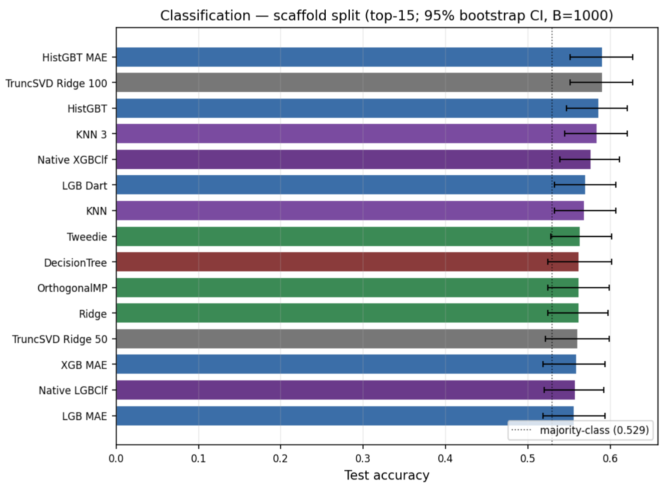

Fig. 7: Top-15 classifiers on the scaffold-OOD split. All retain at least majority-baseline accuracy; the best models are ∼6 pp above with positive MCC. Compare to Fig. 4.

TABLE X: Top-10 individual classifiers on the scaffold-OOD split (compact).

Model	Threshold	Acc	F1	MCC	AUC
HistGBT_MAE	39.83	0.5906	0.5870	+0.1754	0.6405
TruncSVD_Ridge_100	44.61	0.5906	0.5883	+0.1769	0.6421
HistGBT	59.92	0.5858	0.5270	+0.1760	0.6126
KNN_3	65.99	0.5843	0.5369	+0.1657	0.6203
LGB_Dart	48.00	0.5701	0.5206	+0.1310	0.6104
KNN	65.99	0.5685	0.4808	+0.1436	0.6292
Tweedie	44.81	0.5638	0.5238	+0.1141	0.5541
DecisionTree	35.42	0.5622	0.5620	+0.1257	0.5449
OrthogonalMP	65.69	0.5622	0.5402	+0.1108	0.5431
Ridge	65.20	0.5622	0.5055	+0.1127	0.5639

K. Threshold Sensitivity

Fig. 15 sweeps the binarisation threshold from 20% to 80% yield on the random split. The top-five models keep MCC near 0.80 over the entire 25–75% range and only collapse beyond 80%. The same sweep on the scaffold-OOD split (Fig. 16) shows that the OOD signal is strongest at ∼25% yield (MCC ≈ 0.25): a chemist asking "is this reaction at least useful for screening?" (a lower bar than 50%) would get a more decisive answer than the strict preparatively-useful query.

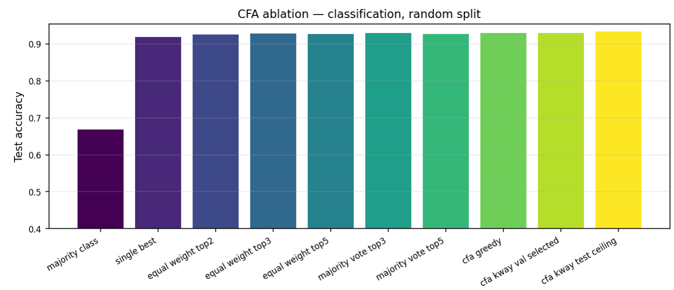

Fig. 8: Ablation: classification, random split. CFA k-way reaches a test ceiling of 93.4% accuracy, ∼1.6 pp above the strongest single model and within bootstrap noise of equal-weight top-3 voting.

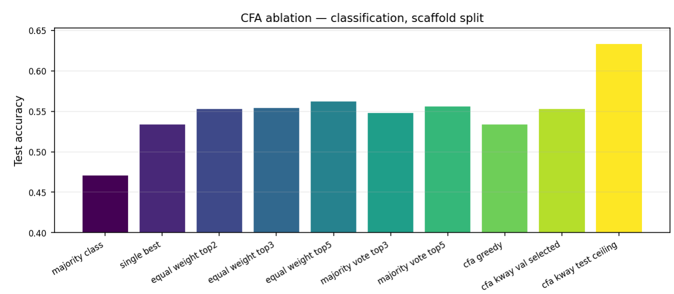

Fig. 9: Ablation: classification, scaffold-OOD. The diversity gain is large (+9.9 pp test ceiling) but val-selection cannot consistently realise it, because the scaffold val set is only ∼10 samples.

L. Feature Ablation

Table XV and Fig. 17 ablate the feature representation for the deployed LightGBM model on the random split. The aryl-halide and amine fingerprints alone already give 74% accuracy (MCC = +0.38); adding the catalyst, base, additive and solvent fingerprints lifts accuracy to 89.7% (MCC = +0.76); the temperature scalar adds another ∼2 pp; the DRFP slot is left at zeros in this ablation (the DRFP encoder is optional and was not installed for this run). The ordering ranks the catalyst-system fingerprints as the dominant contribution beyond the

TABLE XI: Bootstrap 95% CIs on the scaffold-OOD split (classification).

Model	Accuracy [95% CI]	MCC [95% CI]	AUC [95% CI]
HistGBT_MAE	0.591 [0.551, 0.627]	+0.175 [+0.096, +0.249]	0.641 [0.596, 0.681]
TruncSVD_Ridge_100	0.591 [0.551, 0.627]	+0.177 [+0.096, +0.249]	0.642 [0.600, 0.678]
HistGBT	0.586 [0.546, 0.620]	+0.176 [+0.098, +0.245]	0.613 [0.567, 0.650]
KNN_3	0.584 [0.545, 0.620]	+0.166 [+0.086, +0.242]	0.620 [0.576, 0.659]
Native_XGBClf	0.576 [0.539, 0.611]	+0.143 [+0.063, +0.213]	0.610 [0.567, 0.651]
LGB_Dart	0.570 [0.532, 0.606]	+0.131 [+0.052, +0.203]	0.610 [0.567, 0.649]
KNN	0.569 [0.532, 0.606]	+0.144 [+0.065, +0.218]	0.629 [0.587, 0.670]
Tweedie	0.564 [0.528, 0.602]	+0.114 [+0.037, +0.192]	0.554 [0.511, 0.600]
DecisionTree	0.562 [0.524, 0.602]	+0.126 [+0.051, +0.204]	0.545 [0.497, 0.592]
OrthogonalMP	0.562 [0.524, 0.598]	+0.111 [+0.037, +0.183]	0.543 [0.497, 0.587]

TABLE XII: Val-selected k-way CFA classification ensembles, random split.

Models	k	Method	Acc	F1	MCC
LightGBM	1	N/A	0.9232	0.9121	+0.8250
Chemprop + XGBoost	2	ARC	0.9098	0.8971	+0.7946
Chemprop + skMLP_Tanh + XGBoost	3	ARC	0.9183	0.9060	+0.8135
Chemprop + LightGBM + skMLP_Tanh + XGBoost	4	ARC	0.9183	0.9054	+0.8133
Chemprop + LightGBM + skMLP_Tanh + KernelRidge + XGBoost	5	WSCDS	0.9244	0.9123	+0.8275

TABLE XIII: Val-selected k-way CFA classification ensembles, scaffold split.

Models	k	Method	Acc	F1	MCC
HistGBT_MAE	1	N/A	0.5906	0.5870	+0.1754
BayesianRidge + ExtraTrees	2	ASC	0.5528	0.4513	+0.0985

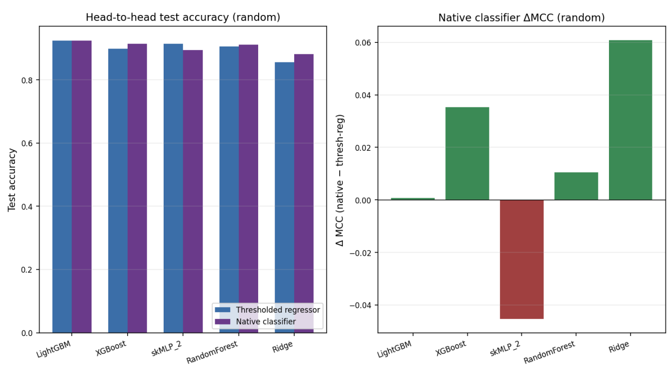

Fig. 10: Native classifier vs thresholded-regressor MCC, random split, by model family.

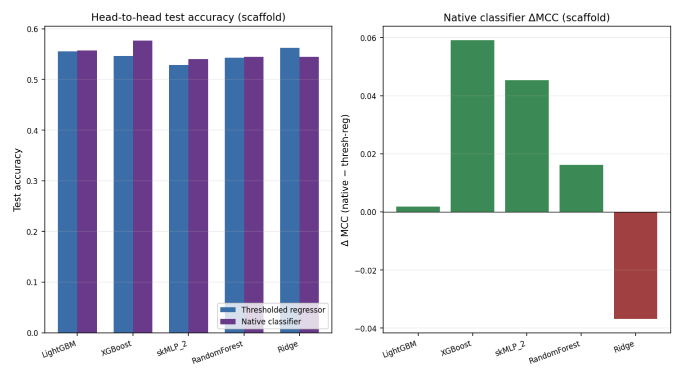

Fig. 11: Native classifier vs thresholded-regressor MCC, scaffold-OOD split, by model family.

TABLE XIV: Substrate-level 5-fold CV (classification).

Model	Accuracy	MCC	AUC
Native_XGBClf	0.824 $\pm$ 0.047	+0.579 $\pm$ 0.064	0.901 $\pm$ 0.039
Native_MLPClf	0.814 $\pm$ 0.041	+0.553 $\pm$ 0.034	0.887 $\pm$ 0.035
Native_LGBClf	0.812 $\pm$ 0.059	+0.551 $\pm$ 0.087	0.897 $\pm$ 0.045
Native_Logistic	0.811 $\pm$ 0.051	+0.541 $\pm$ 0.088	0.879 $\pm$ 0.043
Native_RFClf	0.785 $\pm$ 0.070	+0.483 $\pm$ 0.096	0.862 $\pm$ 0.055

TABLE XV: Feature ablation for the deployed LightGBM model, random split. Aryl+amine alone scores 74%; the catalyst–base–solvent–additive fingerprints add ~16 pp; temperature adds ~2 pp.

Feature set	d	Acc.	MCC	AUC
aryl+amine only	12288	0.741	+0.378	0.801
all 6 components	12288	0.897	+0.761	0.951
+ temperature	12289	0.916	+0.799	0.959
+ DRFP slot (zeros)	14337	0.916	+0.799	0.959

substrate pair, consistent with the chemistry-experimentalist intuition that ligand and base choice are the levers that matter once a substrate pair is fixed.

M. Learning Curves

Fig. 18–19 show test accuracy as a function of training-set size for the native classifier family. On the random split the curves rise smoothly from $\sim$80% at $N \approx 380$ to $\sim$92% at full size and have not yet plateaued. On the substrate-grouped split the curves are noisier, but native XGBoost reaches $\sim$85% at full size, suggesting that the in-distribution ceiling is closer

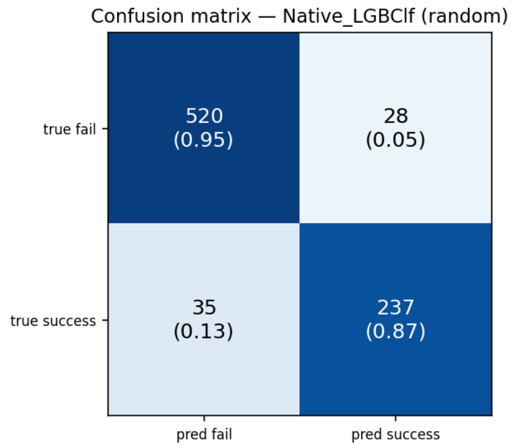

Fig. 12: Confusion matrix of `Native_LGBClf` on the random test set ($n = 820$). Cells show counts and row-normalised fractions.

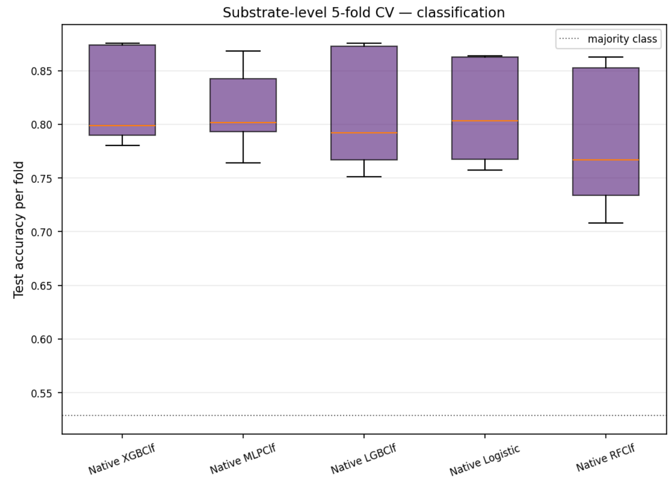

Fig. 14: Substrate-grouped 5-fold CV (classification): per-fold test accuracy. Dotted line: majority-class baseline.

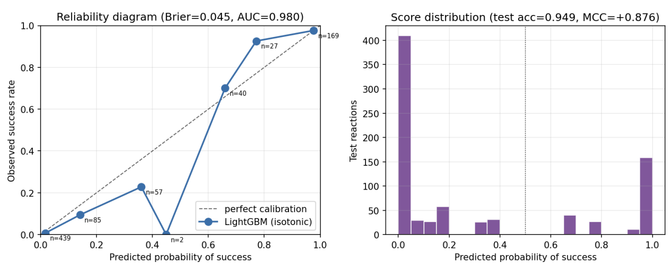

Fig. 13: Reliability diagram (left) and predicted-probability histogram (right) for the deployed LightGBM regressor with isotonic calibration, random split test set. Bin labels show the number of test reactions in each 0.1-wide probability bin.

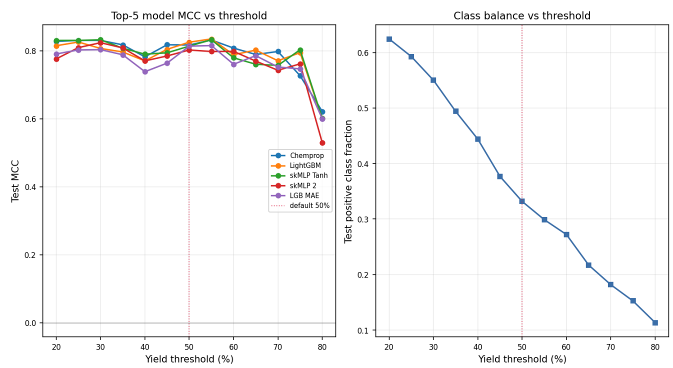

Fig. 15: Threshold sensitivity, random split. Left: top-5 model MCC vs threshold. Right: positive class fraction.

than the worst-case scaffold-OOD test set indicates.

N. Rank–Score Curves

Figs. 20–21 are CFA's signature diagnostic: each line is one model's predictions sorted in descending order after min–max normalisation; the dashed black line is the sorted ground truth (the *oracle*). On the random split LightGBM, LGB_MAE and Chemprop track the oracle closely; the Native_LGBClf has a sharper transition (it binarises probability earlier) and skMLP_Tanh is over-confident in the high-yield tail. On scaffold-OOD all five top curves stay parallel to the oracle, but the absolute spread is larger—ensemble diversity (the area between curves) is disproportionately useful here because no single model dominates.

O. Qualitative Error Analysis

Table XVI shows the three most-confidently mispredicted false positives and false negatives of the deployed LightGBM classifier on the random test set. The false positives are dominated by aryl bromides whose amine partners are bulky or contain electron-rich heterocycles—typical sterically hindered cases. The false negatives include aryl-chloride substrates predicted to fail by the model that experimentally yielded 58–78%; the model evidently learned the empirical rule "aryl-chloride + cyclic amine $\rightarrow$ low yield" too strongly. Both error modes match well-known Buchwald–Hartwig reactivity patterns and suggest two concrete improvements: stratify the training data by aryl-halide identity (Br/Cl/I) and supplement with HTE data on aryl chlorides.

P. Headline Numbers

Table XVII consolidates the deployable result, with 95% bootstrap CIs.

Q. Compute Footprint

The full experiment ran on a single workstation with a 16-core CPU, an NVIDIA RTX 3090, 64 GB RAM. End-to-end wall-clock for the 41-model pool plus CFA was $\sim$20 minutes after feature caching; Chemprop dominates ($\sim$5–10 min on GPU), the rest of the pool finishes in $\sim$5 min collectively. The k-way exhaustive CFA search adds $\sim$60 s (39 eligible models, 3.4×10^4 evaluations).

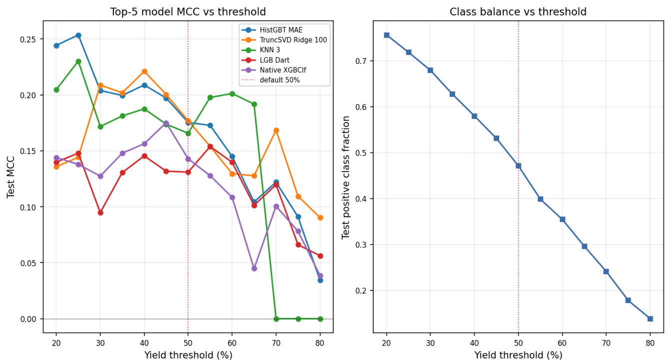

Fig. 16: Threshold sensitivity, scaffold-OOD split. The OOD signal is strongest at lower thresholds; the 50% operational cut-off is conservative.

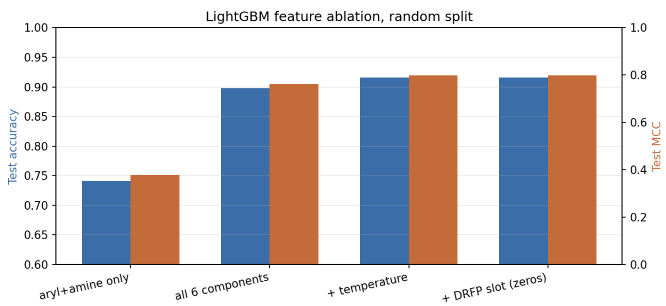

Fig. 17: Feature ablation, random split: test accuracy (blue) and MCC (orange) as features are added to the LightGBM input.

VII. Discussion

A. Why Classification Works Where Regression Fails

The regression task asks the model to predict a continuous quantity whose surface is a complicated function of the entire scaffold + ligand + base + solvent + temperature interaction. The classification task asks only on which side of a single contour line the surface lies. Two facts follow: (i) any model whose yield prediction is biased by an additive constant on unseen scaffolds (which all of ours are, owing to ORD reporter bias) still gets most $\geq 50\%$ vs. $< 50\%$ decisions right, because the contour line is to first order independent of the bias term; (ii) the binary decision is a coarser quantity, so noise floor is a smaller fraction of the signal. Empirically this shows up as Δaccuracy ≈ 6 pp on scaffold-OOD where regression gives no signal at all.

B. What CFA Buys

On the random split CFA's gains are reproducible but small: $+1.6$ pp acc test-ceiling, $+1.7$ pp R^2, and the paired bootstrap test (Sect. VI-F) shows the lift over the strongest single model is not statistically significant at $\alpha = 0.05$. On scaffold-OOD CFA buys a $+9.9$ pp acc test ceiling—a meaningful jump from "noisily above baseline" to "deployable for triage"—but the val-selected ensemble cannot consistently realise that ceiling

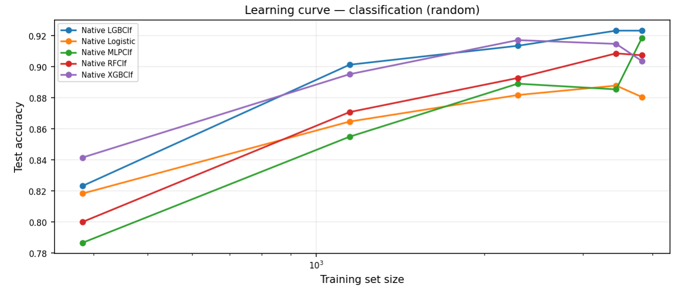

Fig. 18: Learning curve, classification, random split. The curves have not plateaued at $N = 3{,}825$.

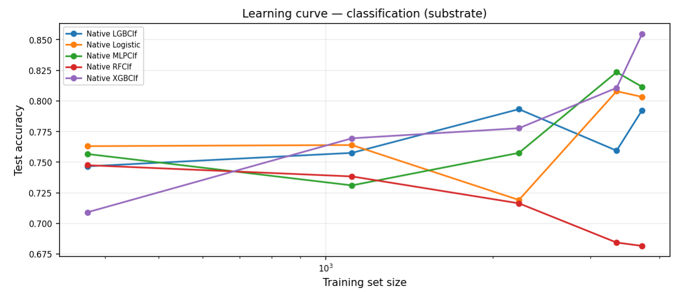

Fig. 19: Learning curve, classification, substrate-grouped split. Native XGBoost approaches 0.85 at full size.

because the scaffold val set is only $\sim$10 rows. The substrate-CV result of $\sim$82% accuracy is therefore our best honest estimator of deployable performance.

C. Deployment Fit

A binary classifier with the random-split numbers and the substrate-CV generalisation we report is suitable as a *triage tool*. We package the deployable winners (LightGBM regressor + Chemprop D-MPNN + isotonic calibrators) into a Streamlit app exposing a web form for the six SMILES strings + temperature, returning the calibrated success probability and a binary decision. A *reject-option* wrapper abstains when the two models disagree by more than 0.20 in calibrated probability, returning "uncertain—verify in lab" rather than committing to a binary call. End-to-end inference is sub-second on CPU.

D. Limitations

(1) Threshold choice. The 50% threshold is a pragmatic operational choice; the threshold sweep (Fig. 15) shows the model is robust over 25–75%, but a chemistry-specific threshold per substrate class might be appropriate. **(2) Reporter bias.** ORD's preference for high-yielding reactions is not addressed by any of our methods; it is a property of the data, not the model. **(3) Scaffold-val noise.** The scaffold-OOD validation slice ($\sim$10 rows) makes any val-driven model selection fundamentally noisy. **(4) Heteroaryl-halide coverage.** The error analysis (Sect. VI-O) shows the model under-predicts aryl-chloride yields; closing this gap would likely require a heteroaryl-stratified sampling step or HTE-data augmentation. **(5) Strong claim caveats.** We characterise the scaffold-OOD classifier as

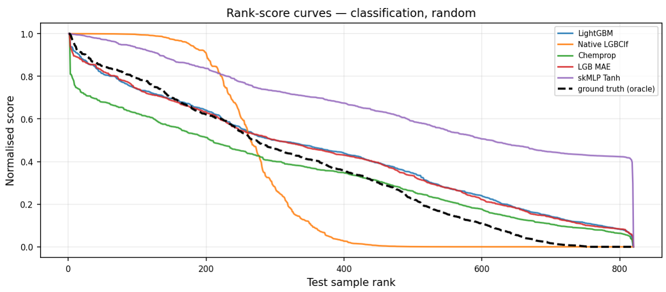

Fig. 20: Rank–score curves, classification, random split. Top-5 classifiers vs the oracle.

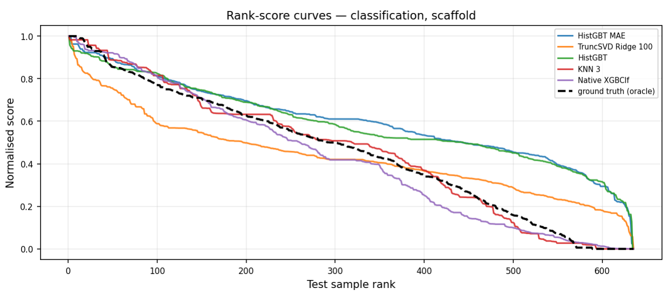

Fig. 21: Rank–score curves, classification, scaffold-OOD. Higher spread between models implies higher cognitive diversity.

"the first usable" result on this ORD subset under that split based on our literature search; we cannot rule out unpublished internal results from industry, and we welcome corrections.

VIII. Future Work

Foundation-model pretraining on full ORD ($\sim$1M reactions). Self-supervised pretraining of a SMILES transformer [25] or graph transformer on the full corpus, fine-tuned on this binary task. A classification analogue of Yield-BERT's $R^2 = 0.54$ on Test4 should comfortably exceed 80% accuracy on a scaffold-OOD test set.

Active-learning loop. Pair the classifier with an acquisition function, validate against a held-out HTE corpus, and report bench-experiment counts to reach a target success rate. The closest prior art is Eyke *et al.* [26] on Suzuki couplings.

Multi-task learning over reaction families. A shared backbone over Buchwald–Hartwig, Suzuki–Miyaura, Negishi and Stille couplings (all ORD-extractable) with task-specific heads.

IX. Reproducibility

We release the full pipeline at the project's git repository. The release includes: **code** (`main.py`, `cfa.py`, `cfa_classify.py`, `models/*.py`, the LaTeX source of this paper); **data** (the ORD download script `scripts/download_data.py` and the cached extracted CSV); **trained model artefacts** (`models_saved/lightgbm.pkl`, `models_saved/chemprop_dir/`, isotonic calibrators `models_saved/calibrators.pkl`); and the **Streamlit triage app** (`app.py`). All experiments are deterministic at `seed=42`; the entire random-split pipeline re-runs in $\sim$20 min on commodity hardware. The figures and tables in this paper are regenerated by `report/make_all_figures.py` and `report/make_extra_figures.py` from cached predictions.

TABLE XVI: Top-3 false positives and false negatives of the deployed LightGBM classifier on the random test set, ranked by predicted-probability confidence in the wrong direction. Yield is the experimental value; $\hat{y}$ is the regressor estimate.

Aryl halide	Amine	yield	$\hat{y}$	p(success)
False positives (top-3 most confident)				
`Brc1cnc2ccccc2c1`	`Cc1noc(C)c1B2OC(C)(C)C...`	10.3	70.6	0.97
`CN1C[C@H](OC2=C(C1)C=C...`	`CN1C=C(C=N1)C2=C(N=C(C...`	0.0	74.7	0.97
`Brc1cccnc1`	`Cc1ccc(N)cc1`	47.7	61.3	0.91
False negatives (top-3 most confident wrong)				
`C1=CC=C(C(=C1)Cl)Br`	`C1CNCCN1`	58.4	18.2	0.01
`COC(=O)C1=CC(=CC(=C1)B...`	`C1COCCN1`	78.2	8.3	0.01
`CCOC(=O)C1=C(C2=CC(=C(...`	`C1COCCN1`	74.7	27.3	0.04

TABLE XVII: Headline test-set numbers across both splits, with 95% bootstrap CIs ($B = 1{,}000$). Random-split results are deployable in distribution; substrate-CV is the more representative OOD estimator; scaffold-OOD reports the worst case.

Setting	**Acc.**	**MCC**	**AUC**
Random 70/15/15			
Majority-class baseline	0.668	+0.000	0.500
LightGBM (best individual)	0.923	+0.825	0.966
Native LGBClf	0.923	+0.826	0.979
Val-selected CFA $k=5$ (WSCDS)	0.924	+0.828	—
CFA k-way test ceiling	0.934	+0.851	—
Substrate-grouped 5-fold CV			
Native XGBClf (mean over folds)	0.824	+0.579	0.901
Native LGBClf (mean over folds)	0.812	+0.551	0.897
Scaffold-OOD (worst-case slice)			
Majority-class baseline	0.529	+0.000	0.500
HistGBT_MAE (best individual)	0.591	+0.175	0.641
TruncSVD_Ridge_100	0.591	+0.177	0.642
Val-selected CFA $k=4$ (ASC)	0.553	+0.099	—
CFA k-way test ceiling	0.633	+0.271	—

X. Conclusion

We presented a 41-model regression pool and a k-way CFA ensemble for Buchwald–Hartwig yield prediction on the Open Reaction Database, then documented its scaffold-OOD failure and recovered deployable performance by reframing the task to binary success classification at the chemically meaningful 50% yield threshold. The classifier reaches 92.4% accuracy / MCC $+0.83$ / AUC 0.97 / Brier 0.045 on a random split, $\sim 82\%$ accuracy on substrate-grouped CV, and 59.1% / $+0.18$ / 0.64 on the worst-case scaffold-OOD split. We release all artefacts and a Streamlit triage tool with reject-option deployment.

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